

The data file is a subsample from the 1982 High School and Beyond Survey and is used extensively in *Hierarchical Linear Models* by Raudenbush and Bryk. The data file consists of 7185 students nested in 160 schools. The outcome variable of interest is student-level math achievement score (**MATHACH**). Variable **SES** is social-economic-status of a student and therefore is a student-level variable. Variable **MEANSES** is the group mean of **SES** and therefore is a school-level variable. Both **SES** and **MEANSES** are centered at the grand mean (they both have means of 0). Variable **SECTOR** is an indicator variable indicating if a school is public or catholic and is therefore a school-level variable. There are 90 public schools (SECTOR=0) and 70 catholic schools (SECTOR=1) in the sample.

```
proc print data=MATHACH (obs=5);
run;
```

Obs	SCHOOL	MINORITY	FEMALE	SES	MATHACH	SIZE	SECTOR	PRACAD
1	1224	0	1	-1.528	5.876	842	0	0.35
2	1224	0	1	-0.588	19.708	842	0	0.35
3	1224	0	0	-0.528	20.349	842	0	0.35
4	1224	0	0	-0.668	8.781	842	0	0.35
5	1224	0	0	-0.158	17.898	842	0	0.35

Obs	DISCLIM	HIMINTY	MEANSES
1	1.597	0	-0.428
2	1.597	0	-0.428
3	1.597	0	-0.428
4	1.597	0	-0.428
5	1.597	0	-0.428

Model 1: Unconditional Means Model

This model is referred as a one-way ANOVA with random effects and is the simplest possible random effect linear model and is discussed in detail by Raudenbush and Bryk. The motivation for this model is the question on how much schools vary in their mean mathematics achievement. In terms of regression equations, we have the following, where $r_{ij} \sim N(0, \sigma^2)$ and $u_{0j} \sim N(0, \tau^2)$,

```
proc mixed data = MATHACH covtest noclprint;
  class school;
  model mathach = / solution;
  random intercept / subject = school;
run;
```

Model Information	
Data Set	WORK.MATHACH
Dependent Variable	MATHACH
Covariance Structure	Variance Components
Subject Effect	SCHOOL
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Containment

Dimensions	
Covariance Parameters	2
Columns in X	1
Columns in Z per Subject	1
Subjects	160
Max Obs per Subject	67

Number of Observations	
Number of Observations Read	7185
Number of Observations Used	7185
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	48102.91726234	
1	2	47116.81230623	0.00000109
2	1	47116.79350024	0.00000000

Convergence criteria met.

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	SCHOOL	8.6097	1.0778	7.99	<.0001
Residual		39.1487	0.6607	59.26	<.0001

Fit Statistics	
-2 Res Log Likelihood	47116.8
AIC (Smaller is Better)	47120.8
AICC (Smaller is Better)	47120.8
BIC (Smaller is Better)	47126.9

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	12.6370	0.2443	159	51.72	<.0001

Model 2: Including Effects of School Level (level 2) Predictors — predicting **mathach** from **meanses**

This model is referred as regression with Means-as-Outcomes by Raudenbush and Bryk. The motivation of this model is the question on if the schools with high **MEANSES** also have high math achievement. In other words, we want to understand why there is a school difference on mathematics achievement. In terms of regression equations, we have the following.

$$\text{MATHACH}_{ij} = \beta_{0j} + r_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{MEANSES}) + u_{0j}$$

Combining the two equations into one by substituting the level-2 equation to level-1 equation, we have

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01}(\text{MEANSES}) + u_{0j} + r_{ij}$$

```
proc mixed data = MATHACH covtest noclprint;
  class school;
  model mathach = meanses / solution ddfm = bw;
  random intercept / subject = school;
run;
```

Model Information	
Data Set	WORK.MATHACH
Dependent Variable	MATHACH
Covariance Structure	Variance Components
Subject Effect	SCHOOL
Estimation Method	REML
Residual Variance Method	Profile

Model Information	
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Dimensions	
Covariance Parameters	2
Columns in X	2
Columns in Z per Subject	1
Subjects	160
Max Obs per Subject	67

Number of Observations	
Number of Observations Read	7185
Number of Observations Used	7185
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	47201.23573408	
1	2	46961.28490236	0.00000000

Convergence criteria met.

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	SCHOOL	2.6357	0.4036	6.53	<.0001
Residual		39.1578	0.6608	59.26	<.0001

Fit Statistics	
-2 Res Log Likelihood	46961.3
AIC (Smaller is Better)	46965.3
AICC (Smaller is Better)	46965.3
BIC (Smaller is Better)	46971.4

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	12.6495	0.1492	158	84.77	<.0001
MEANSES	5.8635	0.3613	158	16.23	<.0001

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
MEANSES	1	158	263.37	<.0001

Model 3: Including Effects of Student-Level Predictors—predicting **mathach** from centered student-level ses, **cses**

$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j} (\text{SES} - \text{MEANSES}) + r_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

Combining the two equations into one by substituting the level-2 equation to level-1 equation, we have

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{10}(\text{SES} - \text{MEANSES}) + u_{0j} + u_{1j}(\text{SES} - \text{MEANSES}) + r_{ij}$$

```
data MATHACH2;
  set MATHACH;
  cses = ses - meanses;
run;
proc mixed data = MATHACH2 noclprint covtest noitprint;
  class school;
  model mathach = cses / solution ddfm = bw notest;
  random intercept cses / subject = school type = un gcorr;
run;
```

Model Information	
Data Set	WORK.MATHACH2
Dependent Variable	MATHACH
Covariance Structure	Unstructured
Subject Effect	SCHOOL
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Dimensions	
Covariance Parameters	4
Columns in X	2
Columns in Z per Subject	2
Subjects	160
Max Obs per Subject	67

Number of Observations	
Number of Observations Read	7185
Number of Observations Used	7185
Number of Observations Not Used	0

Estimated G Correlation Matrix				
Row	Effect	SCHOOL	Col1	Col2
1	Intercept	1224	1.0000	0.02068
2	cses	1224	0.02068	1.0000

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr Z
UN(1,1)	SCHOOL	8.6769	1.0786	8.04	<.0001
UN(2,1)	SCHOOL	0.05075	0.4062	0.12	0.9006
UN(2,2)	SCHOOL	0.6940	0.2808	2.47	0.0067
Residual		36.7006	0.6258	58.65	<.0001

Fit Statistics	
-2 Res Log Likelihood	46714.2
AIC (Smaller is Better)	46722.2
AICC (Smaller is Better)	46722.2
BIC (Smaller is Better)	46734.5

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
3	1065.70	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	12.6493	0.2445	159	51.75	<.0001
cses	2.1932	0.1283	7024	17.10	<.0001

Model 4: Including Both Level-1 and Level-2 Predictors –predicting mathach from meanses, sector, cses and the cross-level interaction of meanses and sector with cses

We have examined the variability of the regression equations across schools. Now we will build an explanatory model to account for the variability. That is, we want to model the following:

$$\text{MATHACH}_{ij} = \beta_{0j} + \beta_{1j} (\text{SES} - \text{MEANSES}) + r_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}(\text{MEANSES}) + \gamma_{02}(\text{SECTOR}) + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}(\text{MEANSES}) + \gamma_{12}(\text{SECTOR}) + u_{1j}$$

Combining the two equations into one by substituting the level-2 equation to level-1 equation, we have

$$\text{MATHACH}_{ij} = \gamma_{00} + \gamma_{01}(\text{MEANSES}) + \gamma_{02}(\text{SECTOR}) + \gamma_{10} (\text{SES} - \text{MEANSES}) + \gamma_{11}(\text{MEANSES}) * (\text{SES} - \text{MEANSES}) + \gamma_{12}(\text{SECTOR}) * (\text{SES} - \text{MEANSES}) + u_{0j} + u_{1j}(\text{SES} - \text{MEANSES}) + r_{ij}$$

The questions that we are interested in are:

1. Do MEANSES and SECTOR significantly predict the intercept?
2. Do MEANSES and SECTOR significantly predict the within-school slopes?
3. How much variation in the intercepts and the slopes is explained by MEANSES and SECTOR?

```
proc mixed data = MATHACH2 noclprint covtest noitprint;
  class school;
  model mathach = meanses sector cses meanses*cses sector*cses
    / solution ddfm = bw notest;
  random intercept cses / subject = school type = un;
run;
```

Model Information	
Data Set	WORK.MATHACH2
Dependent Variable	MATHACH
Covariance Structure	Unstructured
Subject Effect	SCHOOL
Estimation Method	REML

Model Information	
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Dimensions	
Covariance Parameters	4
Columns in X	6
Columns in Z per Subject	2
Subjects	160
Max Obs per Subject	67

Number of Observations	
Number of Observations Read	7185
Number of Observations Used	7185
Number of Observations Not Used	0

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr Z
UN(1,1)	SCHOOL	2.3817	0.3717	6.41	<.0001
UN(2,1)	SCHOOL	0.1926	0.2045	0.94	0.3464
UN(2,2)	SCHOOL	0.1014	0.2138	0.47	0.3177
Residual		36.7212	0.6261	58.65	<.0001

Fit Statistics	
-2 Res Log Likelihood	46503.7
AIC (Smaller is Better)	46511.7
AICC (Smaller is Better)	46511.7
BIC (Smaller is Better)	46524.0

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
3	220.57	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	12.1136	0.1988	157	60.93	<.0001
MEANSES	5.3391	0.3693	157	14.46	<.0001
SECTOR	1.2167	0.3064	157	3.97	0.0001
cses	2.9388	0.1551	7022	18.95	<.0001
MEANSES*cses	1.0389	0.2989	7022	3.48	0.0005
SECTOR*cses	-1.6426	0.2398	7022	-6.85	<.0001

```
proc univariate data = hsbc;
```

```
var meanses;
```

```
run;
```

```
/*
```

```
90%                0.523
```

```
75% Q3             0.333
```

```
50% Median         0.038
```

```
25% Q1            -0.317
```

```
10%               -0.579
```

```
5%               -0.696
```

```
1%              -1.043
```

```
0% Min           -1.188
```

```
*/
```

```
data toplot;
```

```
set MATHACH2;
```

```
if meanses <= -0.317 then do;
```

```
ms = -0.317;
```

```
strata = "Low"; end;
```

```
else if meanses >= 0.333 then do;
```

```
ms = 0.333;
```

```
strata = "Hig"; end;
```

```
else do; ms = 0.038; strata = "Med" ; end;
```

```
predicted = 12.1136 + 5.3391*ms + 1.2167*sector + 2.9388*cses +
```

```
1.0389*ms*cses - 1.6426*sector*cses;
```

```
run;
```

```
proc sort data = toplot;
```

```
by strata;
```

```
run;
```

```
goptions reset = all;
```

```
symbol1 v = none i = join c = red ;
```

```
symbol2 v = none i = join c = blue ;
```

```
axis1 order = (-4 to 3 by 1) minor = none label=("Group Centered SES");
```

```
axis2 order = (0 to 22 by 2) minor = none label=(a = 90 "Math Achievement Score");
```

```
proc gplot data = toplot;
```

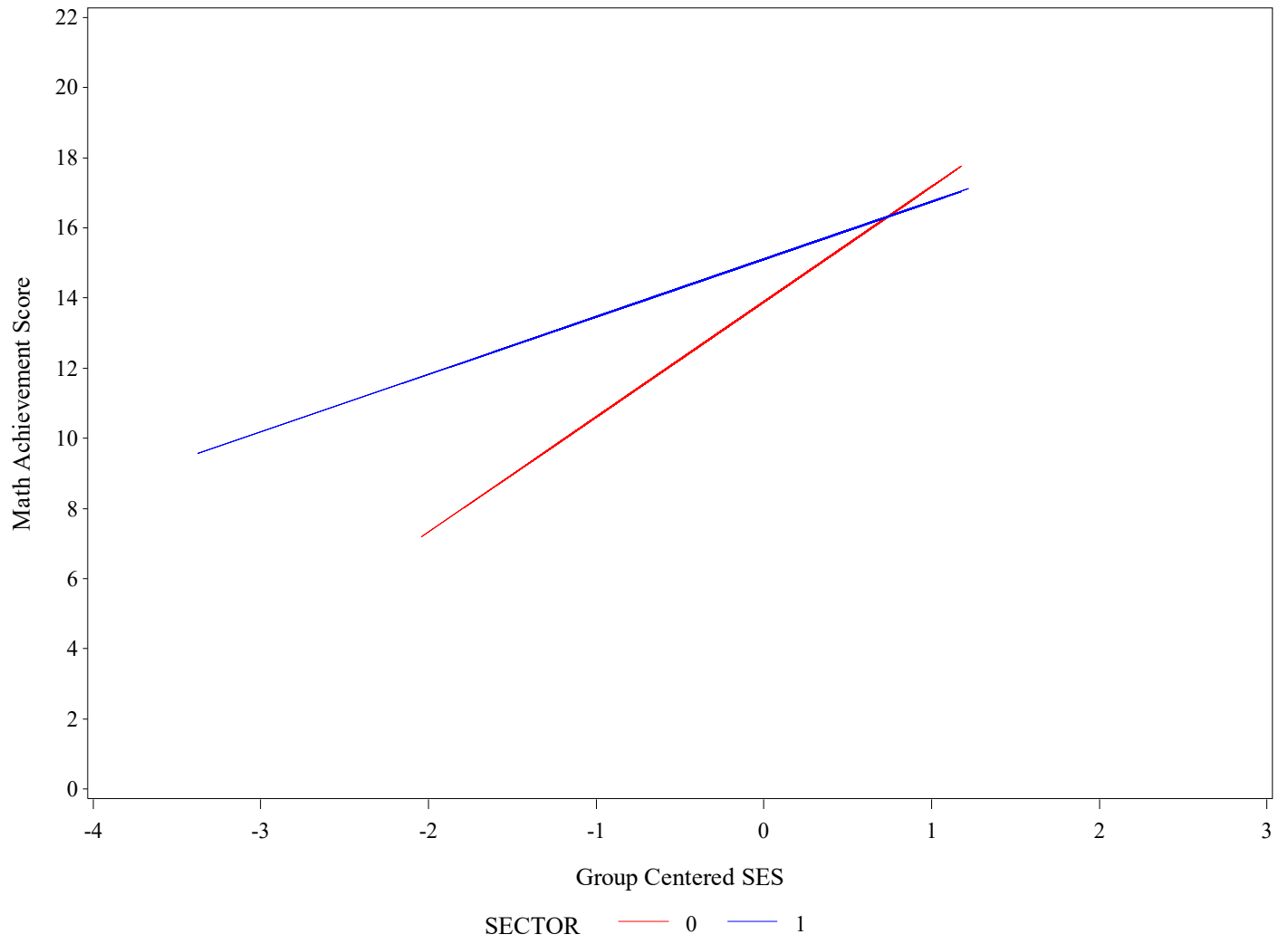
```
by strata;
```

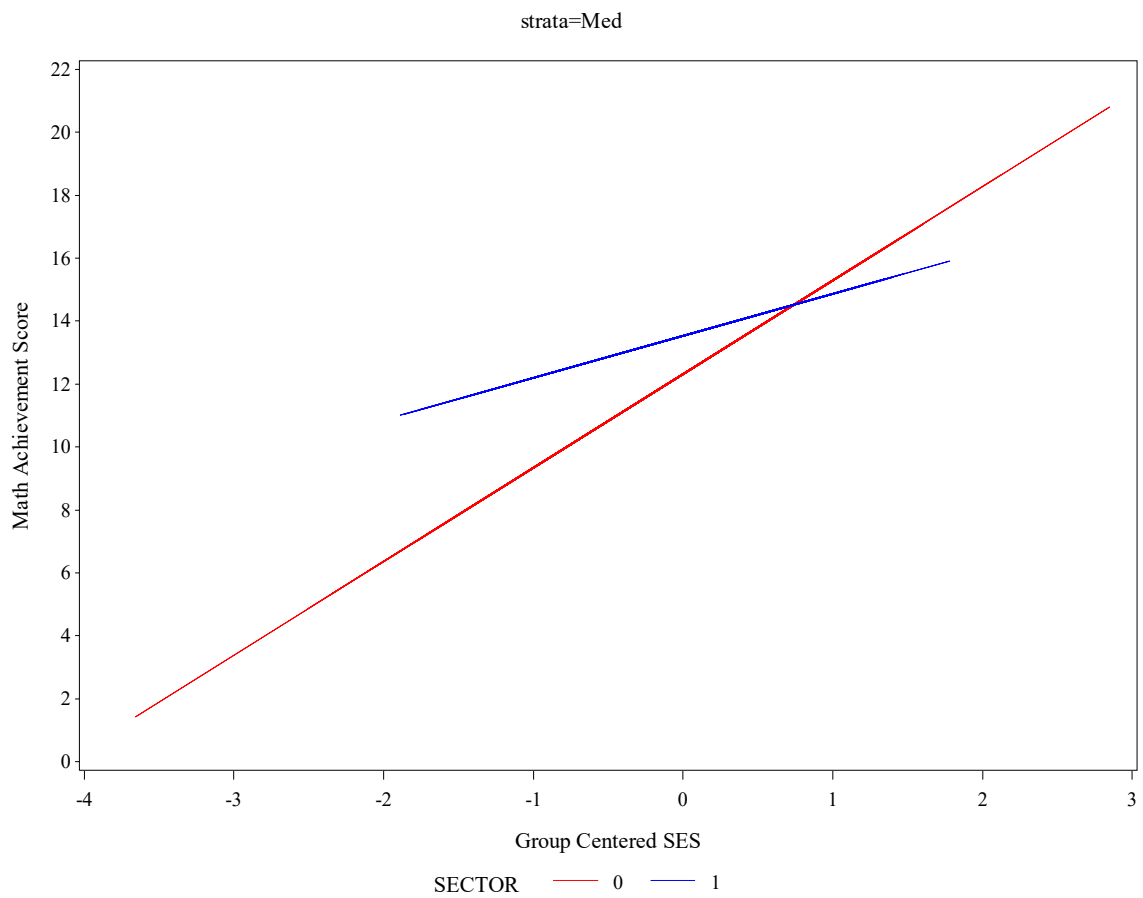
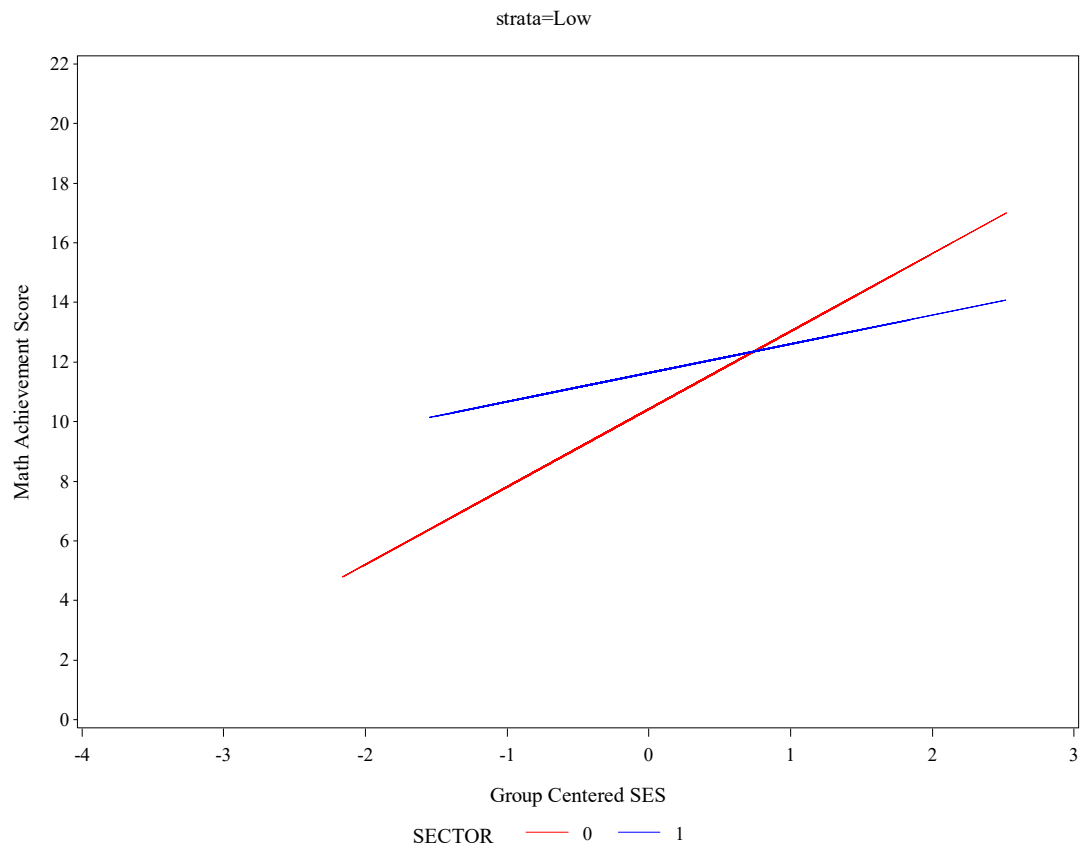
```
plot predicted*cses = sector / vaxis = axis2 haxis = axis1;
```

```
run;
```

```
quit;
```

strata=Hig





Possibly there would be two-way interaction between meanses and sector and a three-way interaction between meanses, cses and sector. We can test it by adding the interaction into the model. For example,

```
proc mixed data = MATHACH2 noclprint covtest noitprint;
  class school;
  model mathach = meanses sector cses meanses*sector
               meanses*cses sector*cses meanses*sector*cses
               / solution ddfm = bw notest;
  random intercept cses / subject = school type = un;
run;
```

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	12.1842	0.2030	156	60.01	<.0001
MEANSES	5.8732	0.5065	156	11.60	<.0001
SECTOR	1.2430	0.3052	156	4.07	<.0001
cses	2.9513	0.1616	7021	18.26	<.0001
MEANSES*SECTOR	-1.1276	0.7355	156	-1.53	0.1273
MEANSES*cses	1.1289	0.4232	7021	2.67	0.0077
SECTOR*cses	-1.6407	0.2406	7021	-6.82	<.0001
MEANSES*SECTOR*cses	-0.1888	0.5997	7021	-0.31	0.7528

Since the variance component for slopes is very small and its corresponding p-value is 0.3177 (not shown). We **cannot** reject the hypothesis that the slopes do not differ across schools. Similarly, we **cannot** reject the hypothesis that the covariance between intercepts and slopes is zero. Therefore, a simpler model can be used:

```
proc mixed data = MATHACH2 noclprint covtest noitprint;
  class school;
  model mathach = meanses sector cses meanses*cses sector*cses / solution ddfm = bw
  notest;
  random intercept / subject = school;
run;
```

Model Information	
Data Set	WORK.MATHACH2
Dependent Variable	MATHACH
Covariance Structure	Variance Components
Subject Effect	SCHOOL
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Dimensions	
Covariance Parameters	2
Columns in X	6
Columns in Z per Subject	1
Subjects	160
Max Obs per Subject	67

Covariance Parameter Estimates					
Cov Parm	Subject	Estimate	Standard Error	Z Value	Pr > Z
Intercept	SCHOOL	2.3752	0.3709	6.40	<.0001
Residual		36.7661	0.6207	59.24	<.0001

Fit Statistics	
-2 Res Log Likelihood	46504.8
AIC (Smaller is Better)	46508.8
AICC (Smaller is Better)	46508.8
BIC (Smaller is Better)	46514.9

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	12.1138	0.1986	157	60.98	<.0001
MEANSES	5.3429	0.3690	157	14.48	<.0001
SECTOR	1.2146	0.3061	157	3.97	0.0001
cses	2.9358	0.1507	7022	19.48	<.0001
MEANSES*cses	1.0441	0.2910	7022	3.59	0.0003
SECTOR*cses	-1.6421	0.2331	7022	-7.04	<.0001

To compare the original model with this simplified one, we can compare their -2LL's, since the fixed portion of these two models are the same.

Model	Number of parameters	-2 LL
restricted	2	46504.8
Unrestricted	4	46503.7

Approximately, the difference in -2LL's is a χ^2 distribution with two degrees of freedom (corresponding to the difference in the number of parameters). The p-value is .577. This justifies the use of the simpler model. The SAS program is shown below.

```
data pvalue;
  df = 2; chisq = 46504.8 - 46503.7;
  pvalue = 1 - probchi(chisq, df);
```

```
run;  
proc print data = pvalue noobs;  
run;
```

df	chisq	pvalue
2	1.1	0.57695